The required modules for running Node.js with RabbitMQ:

```
"amqplib"
"express"
"express-validator"
"mongoose"
```

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To connect to RabbitMQ and create the channel:

```
import amqp from 'amqplib'


const router = Router()

let channel, connection;

async function connectToRabbitMQ(retries = 3) {
  while (retries) {
    try {
      connection = await amqp.connect(process.env.RABBITMQ_URL)
      channel = await connection.createChannel()
      await channel.assertQueue("task_created", {
        durable: true  // This ensures messages persist
      })

      console.log("Connection To RabbitMQ Successfully")

      return { connection, channel }
    } catch (e) {
      console.error("Error In RabbitMQ Connection: " + e)
      retries -= 1
      if (retries <= 0) {
        throw new Error("RabbitMQ Connection Failed")
      }
    }
  }
}

connectToRabbitMQ().then(() => { }).catch(e => process.exit(1))
```

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To send the data to the created channel:

```
router.post('/', createTaskValidation, handleValidationErrors, async (req, res)
=> {
  try {
    const { name, description, user_id } = req.body;
    let response = await fetch(`http://user-
service:3000/api/users/${user_id}/exists`)
    let tempUser = await response.json()

    if (response.status != 200 || tempUser.exists == null) {
      return res.status(400).json({
        'msg': 'Please check the UserId',
      })
    }

    const newTask = await Task.create({
      name,
      description,
      user_id,
    })
    let message = { taskId: newTask._id, user_id, namdid e }

    if (!channel) {
      return res.status(503).json({
        msg: "RabbitMQ channel is not working",
      })
    }

    channel.sendToQueue("task_created", Buffer.from(JSON.stringify(message)), {
persistent: true })
    return res.status(201).json({
      success: true,
      data: newTask,
    })
  } catch (error) {
    // Handle Mongoose validation errors or other DB errors
    res.status(500).json({
      success: false,
      message: 'Server Error',
      error: error.message,
    });
  }
})
```

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To receive from the channel, ex: in the notification service:

```
async function start() {
    try {
        connection = await amqp.connect(process.env.RABBITMQ_URL)
        channel = await connection.createChannel()
        // Assert a durable queue
        const queueInfo = await channel.assertQueue("task_created", {
            durable: true
        })

        console.log(`Queue has ${queueInfo.messageCount} messages waiting`)
        console.log("Connection To RabbitMQ Successfully")
        channel.consume("task_created", (msg) => {
            const taskData = JSON.parse(msg.content.toString())
            console.log("The task data is: ", taskData)
            channel.ack(msg)
        }, {
            noAck: false  // Explicitly set to false for manual acknowledgment
        })
    } catch (e) {
        console.error("Error In RabbitMQ Connection: " + e)
        throw new Error("RabbitMQ Connection Failed")
    }
}

start().then(() => { }).catch(e => {
    console.error("Error In Connection to RABBITMQ: " + process.env.RABBITMQ_URL)
    process.exit(1)
})
```

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